

Exercise – Hit?

A laser source shot Phileas Photon into some direction. Will he stop at an obstacle or travel to infinity? The scenery is considered as seen from above, such that obstacle walls appear as line segments and the trajectory of Phileas is described by a ray.

Input The input file consists of several test cases. Each of them starts with a line containing one integer n ($1 \leq n \leq 20'000$). The following line describes the ray along which Phileas travels. It contains integers x y a b where (x, y) are the coordinates of the starting point of the ray and $(a, b) \neq (x, y)$ is another point on the ray. The following n lines describe an obstacle line segment each. The i -th of these lines contains four integers r s t u , where (r, s) and (t, u) are the endpoints of the i -th segment. All the above coordinates are integers that are in absolute value smaller than 2^{51} . In particular, you cannot represent them using a 32-bit integer data type in general. All numbers on a single line are separated by a single space. The input is terminated by a single line containing 0 (i.e., an empty testcase).

Output The output for each testcase appears on a separate line. The line consists of the word yes, if the ray hits an obstacle¹ and no, otherwise.

Sample Input

```
1
0 0 1 1
2 0 1 2
1
1 1 0 0
0 -2 -1 0
2
0 0 1 1
-1 -2 -1 0
2 0 2 1
2
0 1 1125899906842623 1125899906842623
1 2 1 3
1125899906842621 1125899906842620 1125899906842621 1125899906842621
3
1125899906842623 1125899906842623 0 1
1125899906842621 1125899906842620 1125899906842621 1125899906842621
1 2 1 3
-1 0 -1 1
0
```

Sample Output

```
yes
yes
no
no
yes
```

¹The obstacles segments are relatively closed, that is, both endpoints are included into consideration.